

3(a)	Acceleration / force proportional to displacement (from a fixed point)	1
	either	
	acceleration and displacement in opposite directions	1
	or	
	acceleration always directed towards a fixed point	

3(b)(i)1.	0.10 s, 0.30 s, 0.50 s, etc (any two)	1
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3(b)(i)2.	either 0.0 s, 0.40 s, 0.80 s, 1.20 s or 0.20 s, 0.60 s, 1.00 s (any two)	1
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3(b)(ii)	period = 0.40 s frequency = 2.5 Hz	1
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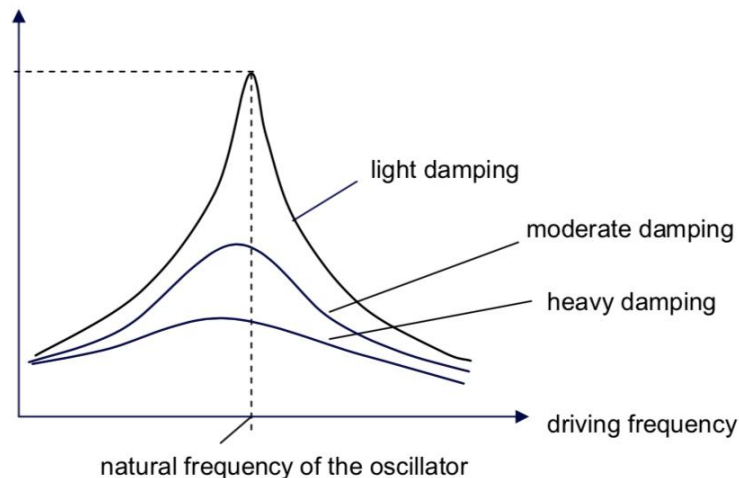
3(b)(iii)	phase difference = 90°	1
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3(c)	Free oscillation: object oscillates at its natural frequency without external forces applied. (without any loss of energy / no resistive forces)	1
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Forced oscillation: object made to oscillates at the frequency of an external periodic driving force. (continuous energy input / driving oscillator)

(d)(i)

Amplitude of forced oscillations



Any of these shapes and max peak at natural frequency

3(d)(ii)	frequency = 2.5 Hz	1
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